Probabilistic Modeling in Design

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A general procedure is proposed for evolving the form of a density function that is consistent with the concept of subjective probability. The procedure directly applies new data information to the updating of the form of a density function without imposing on it any theoretical distribution that could restrict its shape, and permits the direct use of judgment arising from real world experience. It is based on the simple concept that sample size is a measure of confidence in the shape of a density function. Two possible algorithms are given, and the concept is extended for simple "true" or "false" events. The importance of probability in artificial intelligence is also dicusssed, and its essentially subjective nature is described. Procedures are briefly suggested.

1 Introduction

One of the new design techniques made possible by the digital computer is probabilistic design. In this approach, uncertainty about a design parameter or variable is coded in a probability density function. Then probabilistic analysis is used to estimate the probability that a device or system will meet performance requirements. This can be considered a more meaningful approach than using the traditional factor of safety. However, it requires that the designer also generate somehow, an acceptable level of probability of satisfactory performance. The design criterion for feasibility now is

Predicted probability of performance
≥ accceptable probability of performance

Our concern in this paper will be restricted to the primary problem of codifying uncertainty about the primary parameters and design variables into probability density functions. To obtain a full appreciation of this problem it is desirable to see its role in the overall probabilistic design process; and an overview of this is given in the next section.

Central to our whole approach is the concept of *subjective probability*, and we shall argue that by the nature of the engineering design process, probability must be considered inherently subjective. Codification techniques for density functions must be consistent with this definition of probability and arise naturally out of it.

Artificial intelligence shows considerable promise as a technique for automating and improving the design process, in particular in expert systems. And again this technique must be founded on subjectively based probability distributions, which are used as the basis for risk judgments, an essential component of applying expertise to design.

2 An Overview of Probabilistic Design

As a procedure, probabilistic design has several components. The first is the problem of generating probability density functions from known information, which may be in the form of numerical data, or simply as general experience in the field. There are several techniques for doing this.

The best known, and the most commonly used by mathematicians, is perhaps the use of *statistical parameter estimation* techniques, in which the parameters associated with assumed analytical distributions, such as the normal, the Weibull, the exponential, and so on, are estimated using recorded data, or measurements of the variable in question.

A common technique in engineering work is *mean rank plotting*. If a sample of values for a random variable is ordered from high to low, then the ith member can be thought of as a random variable having different values for every sample of the same size obtained. It is called the ith order statistic, and the corresponding value of the cumulative distribution function of the primary variable is called the ith rank. It can be shown that the mean rank is given by

$$\xi = i/(n+1) \tag{1}$$

This can be used as an estimate of the value of the cumulative distribution function for the ith ordered value of the sample. Some theoretical cumulative distribution functions can be linearized by variable transformations so that a linear plot is possible. The transformed sample values are plotted against the transformed mean rank value, and a least squared error fit is used to determine the parameters of the distribution. The Weibull distribution is commonly used in this way in reliability work; and the convenience of this procedure perhaps accounts for its great popularity in this application.

A very powerful and general procedure for generating a probability density function is based on Jayne's principle: *The minimally prejudiced probability distribution is that which maximizes the entropy, subject to constraints supplied by the given information*. The most convenient form of the information is the moments of the distribution calculated from the sample values. Very good fits can be obtained using five or six moments [1], and the density function has the convenient form

$$f(x) = \exp(a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n) \tag{2}$$

where n equals the number of moments used.

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If we accept the idea that probability is a strictly subjective concept when applied to design, then the engineer is quite justified in working with probability distributions even when no sample values are available. In this event a density function represents a codification of the engineer's judgment of the random nature of a quantity. In fact, to be realistic we must concede that there is a subjective element in the creation of any density function, even if hard data are available and used in the previous methods.

The second component of probabilistic design is probabilistic analysis, in which we wish to determine the random nature of a dependent variable y, that is a known function $g(x_1, x_2, \ldots, x_m)$ of a set of independent variables $x_1, x_2, \ldots, x_m$, and the random nature of the x's has been defined. There are several methods of doing this, the best known and most powerful of which is the Monte Carlo method [1].

The third central component of probabilistic design is the prediction of failure mode probabilities. Here here we are concerned with predicting the probabilities of the following kinds of events being true or false.

$$y_i(x) - S_i \geq 0 \tag{3}$$

$$y_i(x) - S_i \leq 0 \tag{4}$$

where x is the set of independent variables and parameters describing the design, $y_i(x)$ is an expression or set of expressions that predicts the value of a performance characteristic, and S_i is a specification. These expressions define the feasibility of the design, and their associated probabilities define the risk that it will not perform satisfactorily. These probabilities can be estimated by using the methods of probabilistic analysis. This topic in probabilistic design encompasses all of reliability theory, and an extension of it is probabilistic optimization, sometimes called stochastic programming.

Other important topics in probabilistic design are the probabilistic analysis of sequential events, order statistics, the theory of extreme values, and the design option problem in which the designer is choosing the best design from two or more optional designs.

3 The Problem of Coding Probability Density Functions Using Limited and Evolving Knowledge

There is a natural way to develop probability density functions for engineering type random variables, based on a rational combination of subjective judgment and gradually accumulated information in the form of either explicitly observed sample values, or general background information. When the engineer is first confronted with the problem of estimating the form of a density function, he or she may have no observed sample values at all, but will always have some general information about the general random nature of the variable, even if very limited. This will usually be based on experience, either direct or acquired, with similar variables in somewhat similar devices or systems. Judgment should be used to formulate a preliminary density function. This curve can be adjusted in the light of additional information, preferably but not necessarily in the form of specific observed values. As more of this information becomes available, the density function can be *evolved* to a better and better estimate of its shape.

Perhaps the best known method for attempting to predict density functions using evolving knowledge is the use of Bayes' Theorem. I have argued elsewhere [1, 2] that this, on several grounds, is really not a rational procedure in design. First, theoretical distributions must be arbitrarily chosen to represent both the primary random variable, and also the parameters of this primary distribution. The second difficulty is that the choice of the distributions for the parameters is supposed to be subjective, and the basis for updating them is the combining of new data with a form of Bayes' equation. It is difficult to conceive how engineering judgment can be brought to bear on the initial distribution of the parameters, since they have no relationship to real world experience. And the final difficulty is that almost no theory exists for applying this method simultaneously to more than one distribution parameter.

What is really needed is a procedure that directly applies new data information to the updating of the form of a density function without imposing on it any theoretical distribution that could restrict its shape, and that permits the direct use of judgment arising from real world experience.

In earlier publications I have described an evolutionary method that begins to address this problem [1, 2]. It is based on the concept of evolutionary stochastic similarity between an existing and relatively well-known variable in the random sense, and a new random variable. The hypothesis is that, if the modeling is essentially the same in some sense, and the environmental loadings are essentially similar, then the *form* of the probability function will be unchanged, and that only the *scale* will change. This idea can be used to derive a new density function, using new data combined with the known random variable's density function.

4 The General Evolutionary Method

The abovementioned evolutionary approach would seem to

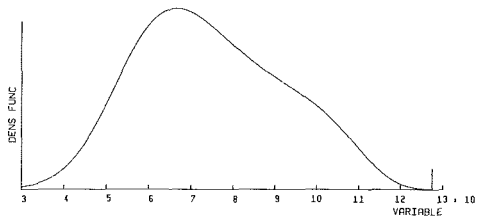


Fig. 1 Subjective density function used for simulation

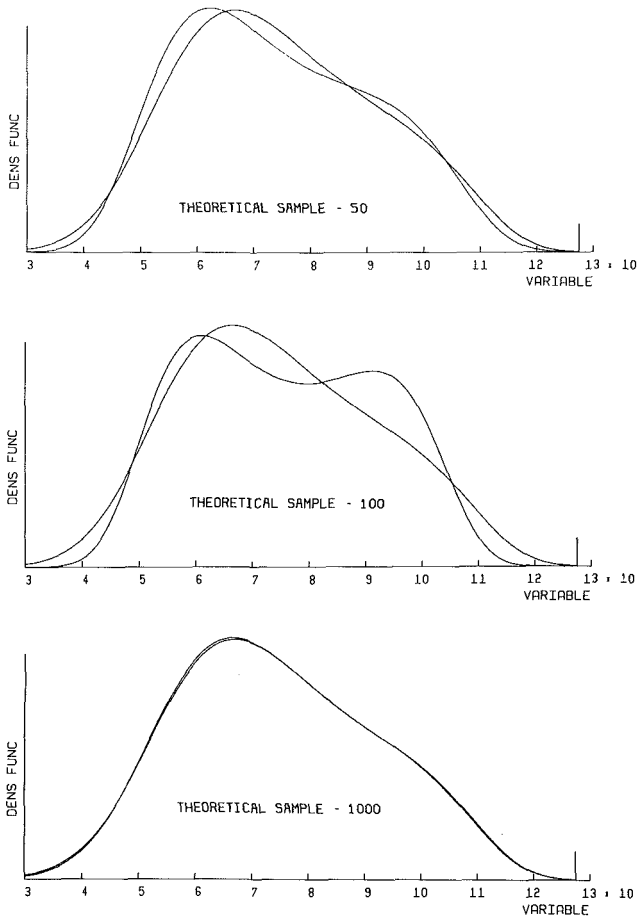


Fig. 2 Reconstruction of the simulated subjective density function, using different theoretical sample sizes

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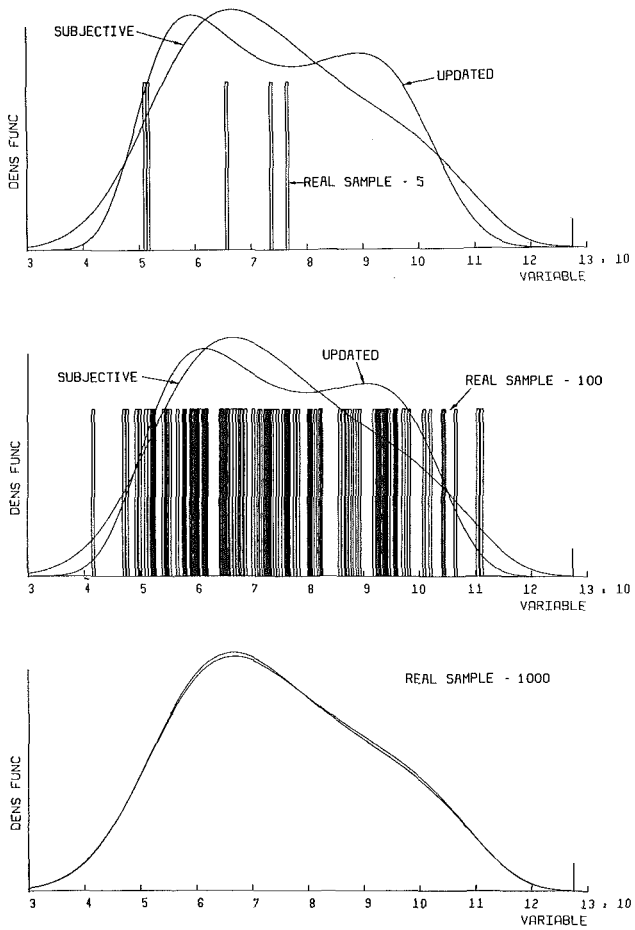


Fig. 3 Subjective and real functions are identical; subjective sample size 100.

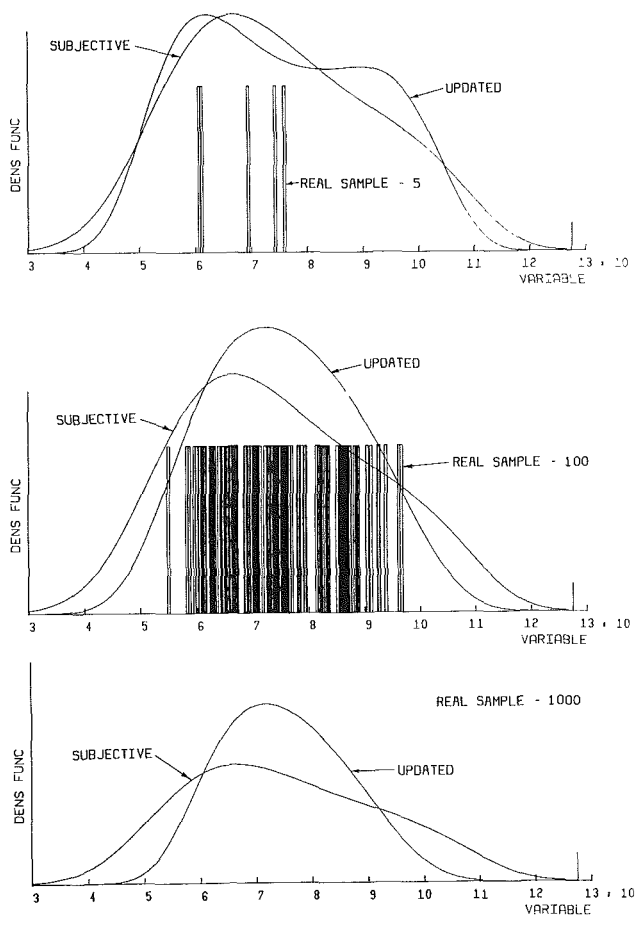


Fig. 4 Real function has 0.6 scale and the same mean as the subjective function; subjective sample size 100.

be a useful one, since very many new random variables in design are closely related to some previous variable that has its random nature well known. However, it would be desirable to have an evolutionary approach that is completely general, and based on general intuitively applied knowledge, rather than knowledge related to any specific existing random variable. Such a method is proposed in the following algorithm. It is based on the simple concept that *sample size* is a measure of confidence in the shape of a density function. It is a sort of inversion of the classical statistical concept of determining a confidence level that is a function of sample size. But the procedure is not in any sense related.

Algorithm 1 • General Evolutionary Method

1 A density function is subjectively sketched.
2 Numerical coordinates are obtained in order to discretely define the function.
3 The first four or five central moments are calculated.
4 A maximum entropy density function is fitted to the curve, using the moments.
5 A theoretical sample is generated from this density function using the rejection method [1]. The theoretical sample size will correspond to the confidence in, or the extent of, the knowledge base for the initial subjectively sketched curve. This choice of sample size must, of course, be subjective, and will occupy a range indicated below.

0 = no knowledge or experience
•
•
50 = roughly defined
•
•
1000 = fairly good confidence
•
•
10,000 = very accurately defined

Research will be required to define more accurately this scale.

6 Any new data based on observations are combined with the theoretical sample, and used to calculate central moments. However, if this real sample is small (< 10), then only the mean is changed in the moments of the subjective distribution.
7 A new maximum entropy function is generated, based on these moments, and used to represent the current state of combined judgment and data in the shape of the density function.
8 The engineer can at any time subjectively alter the current density function, based on new general experience or information, perhaps using advice from colleagues.
9 And any new data can be injected into the reconfiguring of the density function by starting again at the beginning of the algorithm.

Thus the use of a theoretical sample from the prior subjective distribution makes it possible to *blend* the subjectively based information with the hard data. The blend is then used to generate an updated or *evolved* density function.

This procedure enables a density function to be generated in a situation where the engineer initially has no sample data, but can make some subjective estimate of the density function based on general experience with similar random variables. This initial estimate can then be updated by using any measured or observed values that become available. Although the method

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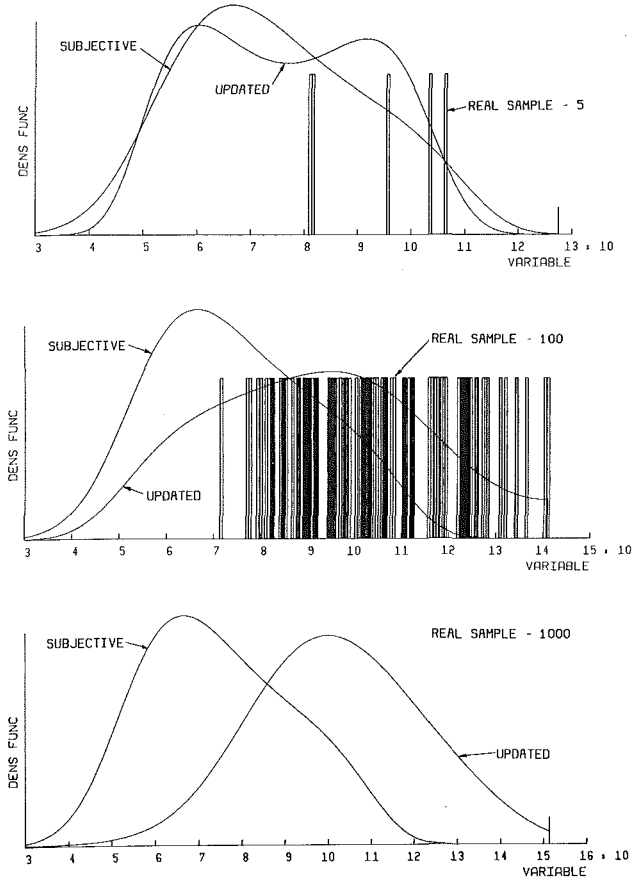


Fig. 5 Real function is shifted right 30 units; subjective sample size 100.

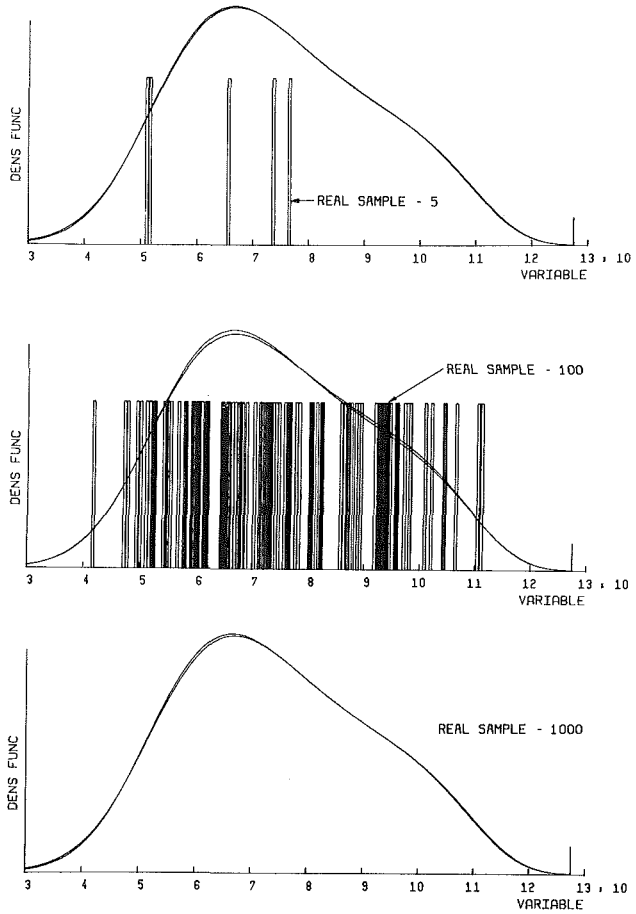


Fig. 6 Subjective and real samples are identical; subjective sample size 1000.

has some similarity in concept to the use of Bayes' equation, referred to earlier, it does not require that any analytical distribution except the unbiased maximum entropy function be imposed on the random variable. Of even more importance, it does not require the use of arbitrary analytical distributions for the parameters of the parent distribution. In this method the parent distribution itself is directly taken as subjective, whereas in Bayes' equation the distribution of the parameters is taken as subjective. Judgment can therefore be much more easily applied to the problem. And finally, there are no mathematical limitations on the scope of the method.

It is quite difficult to verify a procedure such as this by experimentation. However, a *simulation* of the method is possible, which will give some insight into how it works and how large the theoretical sample should be. An arbitrarily selected subjective distribution is shown in Fig. 1. It is a relatively smooth curve with one rather minor kink in it to illustrate sensitivity to change. In order to isolate the effects of differences between the "real" distribution and the subjective one, the "real" distribution was made exactly the same in shape in the first trial, then scaled to 0.6 with the same mean in the second trial, and finally only shifted by plus 30 units. The "real" sample was obtained by generating a theoretical sample from the "real" distribution, using again the rejection method.

Trials were made for each of the "real" distributions, for all combinations of the following: a subjective sample size of 10 and 1000, and a "real" sample size of 5, 100, and 1000. The plotted results are shown in Figs. 2–8. Each member of the "real" sample is plotted as a vertical line on the figures, except for a sample size of 1000.

Figure 2 simply illustrates the relative accuracy with which the curve is recreated from different theoretical samples sizes, using the maximum entropy distribution. It gives an indication of how sensitive the shape is to sample size. A sample size of 50 or 100 can give considerable discrepancy.

Figures 3, 4, and 5 give the simulated results when the subjective curve is represented by a theoretical sample size of 100, and sample sizes of 5, 100, and 1000 are simulated from each of the three "real" curves. Figures 6, 7, and 8 represent the same update simulation for a theoretical sample size of 1000.

In Fig. 3 the subjective and real density functions are identical. The theoretical sample size is 100, and the real sample varies from 5 to 1000. A real sample of 5, or even of 50, can give a surprisingly large false change to the updated curve. With 1000, the curve is almost unchanged.

Figure 4 is the same, except the real function is scaled to 0.6 of the subjective one. The real samples used in each figure are identical, since they are pseudo-random. As the real sample increases we get the expected trend toward the real function. Again the sample of 5 gives a false distortion.

The real function is shifted 30 units in Fig. 5, and the results are similar. The engineer would likely increase the upper bound of the updated function.

Figures 6, 7, and 8 are a repeat of the previous three, but simulate a subjective function in which the engineer has more confidence; they are represented by a theoretical sample size of 1000. As a result, there is less change in the updated functions when the real sample is incorporated.

The previous figures did not, in fact, incorporate the feature in the algorithm requiring that only the mean, or first moment, would be affected by a real sample size of less than 5. Figure 9 incorporates this feature for a sample size of 5 in the three real functions, and how the false distortion has been eliminated.

Unbiased distributions can also be represented by histograms, created by arbitrarily discretizing the random variable into equally spaced intervals. This leads to the following alternative algorithm.

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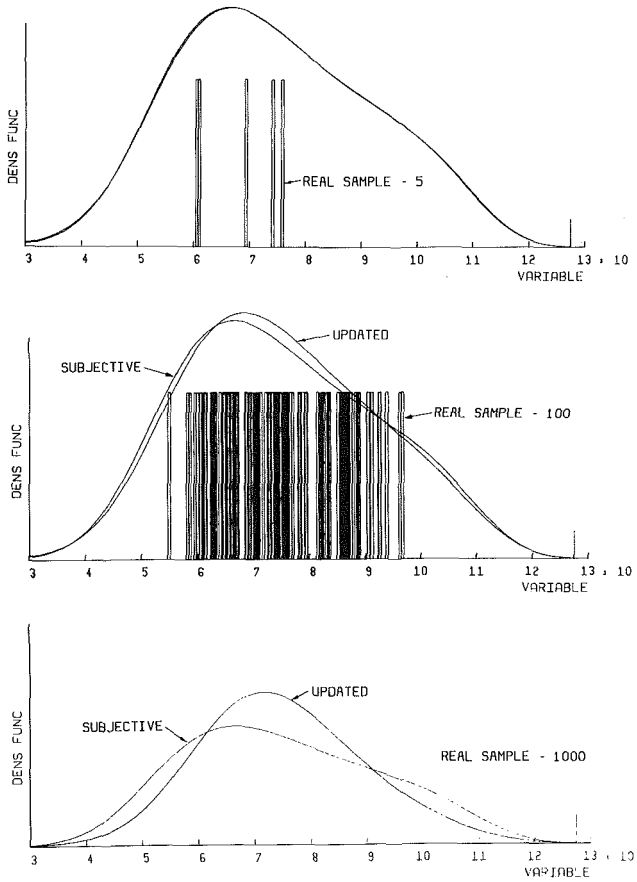


Fig. 7 Real function has 0.6 scale and the same mean as the objective function; sample size 1000.

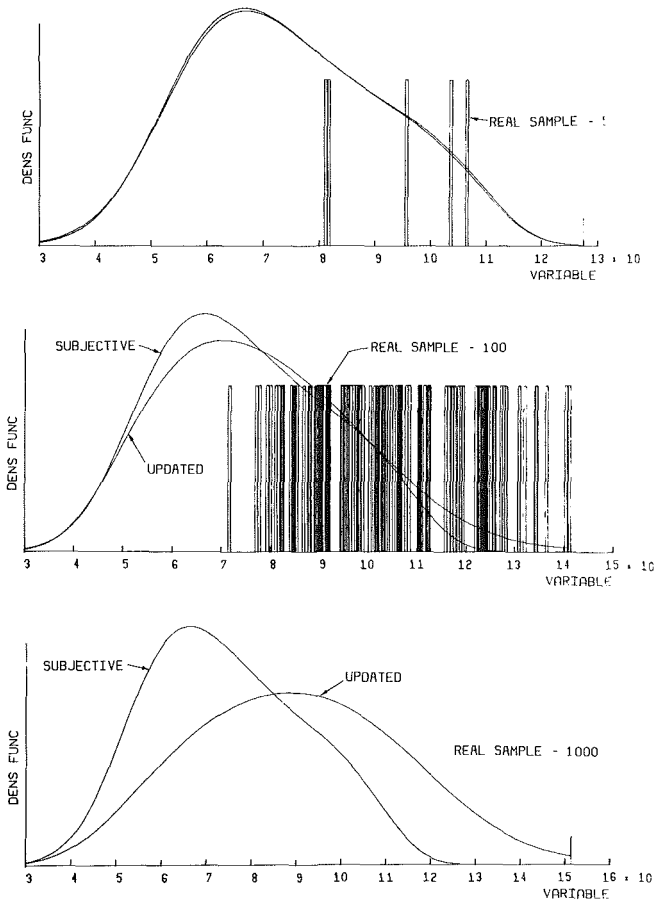


Fig. 8 Real function is shifted right 30 units; subjective sample size 1000.

Algorithm 2 • Histogram Method

1 A density function is subjectively sketched. It may be convenient to fit a maximum entropy distribution to it.
2 The variable is arbitrarily discretized into equally spaced intervals.
3 A theoretical sample size, n_s, is selected in the same manner as in Algorithm 1.
4 The area under the subjective density function is determined for each interval by integrating over the interval, designated A_i.
5 Each interval is assigned a *subjective frequency* by the relationship

$$f_i = A_i n_s \tag{7}$$

6 Any new data based on observations are assigned to the appropriate histogram interval and combined with the subjective frequencies.
7 This yields a new histogram which can be converted to a density function by curve fitting and normalizing the resulting curve.

This algorithm is simpler than Algorithm 1, but is overly sensitive to the local effects of small sample sizes. Some numerical smoothing is possible, but it does not give the elegantly smoothed and fitted curves of Algorithm 1.

A second situation can exist in design work, in which we are concerned with discrete events that are either "true" or "false," rather than with continuous random variables. We now wish to evolve an estimate of the probability that the event will be "true" or "false," or "will occur" or "will not occur." A similar approach can be used. Again we begin with a subjective judgment of the single probability, and then update it in the light of observed occurrences. And again we shall use the concept of representing the level of our confidence in a subjective estimate by the equivalent subjective sample size. The use of the terminology "level of confidence" seems unavoidable here, but should not be confused with the use of the term in classical statistical theory, to which it has no correspondence.

Using this concept, which is really a special case of the histogram algorithm described above for random variables, we can write the following update equation.

$$p_u = \frac{p_s n_s + f n_r}{n_s + n_r} \tag{8}$$

where

p_u = updated probability,
p_s = subjective probability,
n_s = subjective sample size,
f = fraction of the observed sample being "true,"
n_r = observed sample size.

Bayes' equation is also commonly used with discrete events, but it always seems to require unnecessary complication and obscurity.

5 Artificial Intelligence

Human decision making is primarily intuitive, because it is based on incomplete and unformulated information; and much of current artificial intelligence applications is the setting up of programmed intuitive decisions to handle particular types of situations. This would seem particularly true of the two major engineering applications of artificial intelligence, machine vision and expert systems. Because of incomplete information and associated uncertainty, such decision making is inherently probabilistic, and it would seem unrealistic not to incorporate a probabilistic approach in these fields. Because little or no hard, measurable data is available on which to base

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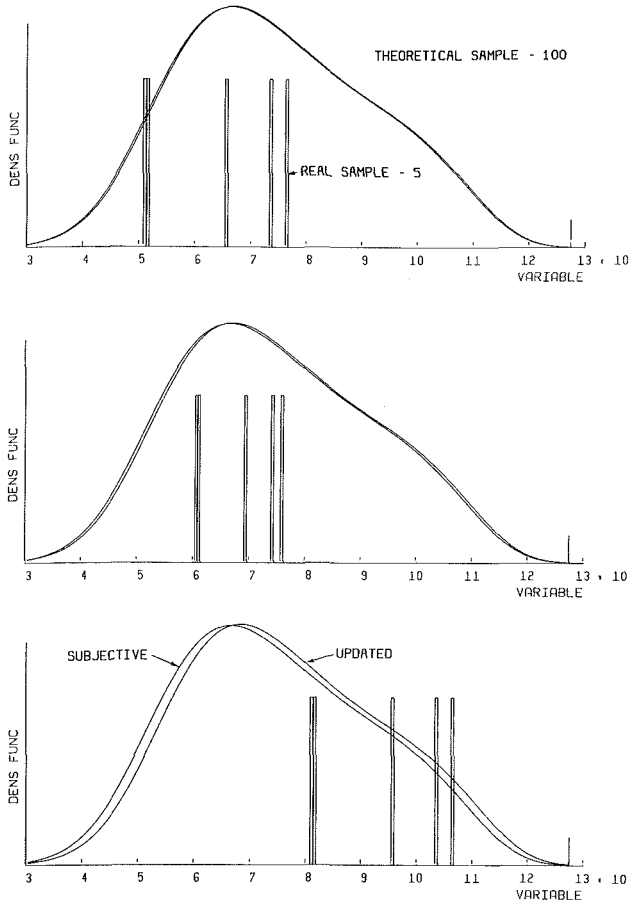


Fig. 9 Effect for the three real functions when only the first moment is changed

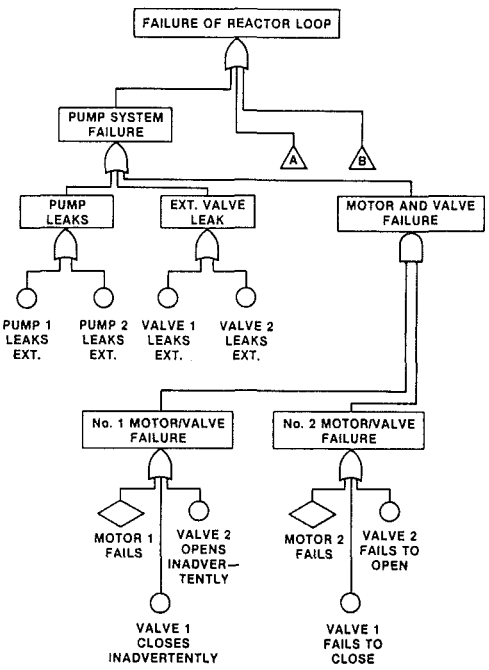


Fig. 10 Fault tree diagram for a reactor

probability estimates, it would seem also true that they must be subjectively based.

In expert systems, a *rule* is simply an IF/THEN/ELSE algorithmic structure. In its simplest form it can be expressed as

IF E THEN H

where E is evidence or an observed event that is true or false, and H is a hypothesis or action that is either true and should be followed, or should be ignored. These structures are assembled in networks, using a real expert's knowledge, so that for a specific application in the area of expertize encompassed by the software, the input of requirements and relevant data is interpreted by the rules to give a recommended solution. There is always some uncertainty in the expert's judgment about the formulation of a rule, and so a probability should be associated with it. Computer scientists who have developed expert systems seem reluctant to use a probabilistic approach, perhaps because of an unwillingness to accept its inherently subjective nature. In a rare example when uncertainty was recognized, Shortcliffe [3], in the MYCIN expert system for medical treatment, used an *uncertainty factor* ranging from -1 to $+1$, as a measure of uncertainty. This was presumably done to avoid the need to accept anything called *probability* as being subjective.

Expert system networks are similar to reliability fault trees, which are represented by Boolean logic gates as shown in the example in Fig. 10. In fact such a tree could be interpreted as an expert system used to decide if a mechanical system will fail or not, particularly if the probabilities are based on the expert's judgment.

Uncertainty can occur at different points in the rule, depending on the type of system. In Fig. 10, it is assumed that if motor 2 fails, *or* valve 1 fails to close, *or* valve 2 fails to open, *then* no. 2 motor/valve will fail for sure. In some systems there could be uncertainty about this conclusion, and a probability would have to be associated with it.

Probabilities can be evolved in expert systems using the procedures described above. It is necessary to propagate probabilities through an expert system, just as it is with fault trees, and this can be done using similar methods. Monte Carlo simulation is always a simple and straightforward way to solve such problems, although sometimes expensive in computer time.

6 Conclusions

Subjective probabilities are pervasive in design practice. As we expand the use of coding uncertainties in design by using probability techniques, it is important that we develop meaningful procedures for combining judgment with hard data, and one possible approach is suggested using the concept that a sample size is a measure of confidence in the shape of a density function.

It is my belief that in many situations the subjective sample size would commonly be 100 or more, and also commonly would have equal or greater significance than the real sample.

Some experience is needed in using this as a tool. We need to become familiar with the practice of using sample size as a *direct* measure of the extent of subjective knowledge.

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